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Electronic device including antenna module

Abstract

A portable electronic device is provided. The portable electronic device includes a metal housing, an antenna module positioned in the metal housing, a first injection-molded member positioned in a radiation direction of the antenna module and having a first dielectric constant, a second injection-molded member at least partially in contact with the first injection-molded member, positioned in the radiation direction of the antenna module, and having a second dielectric constant different from the first dielectric constant, and a first organic film disposed along an inner surface of the metal housing to bond with at least a portion of the first injection-molded member, wherein the second injection-molded member forms, together with the metal housing, an exterior of the portable electronic device, and wherein the second injection-molded member is coupled with the first injection-molded member through a coupling structure in which one of the first injection-molded member or the second injection-molded member is at least partially protruding into another one of the first injection-molded member or the second injection-molded member.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365(c), of an International application No. PCT/KR2022/007212, filed on May 20, 2022, which is based on and claims the benefit of a Korean patent application number 10-2021-0076961, filed on Jun. 14, 2021, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) The disclosure relates to an arrangement and coupling of structures included in an electronic device to enhance waterproof performance and antenna performance.

BACKGROUND ART

(2) Advancing information communication and semiconductor technologies accelerate the spread and use of various electronic devices. In particular, recent electronic devices are being developed to carry out communication while carried on.

(3) The term “electronic device” may mean a device performing a particular function according to its equipped program, such as a home appliance, an electronic scheduler, a portable multimedia player, a mobile communication terminal, a tablet personal computer (PC), a video/sound device, a desktop PC or laptop computer, a navigation for automobile, etc. For example, electronic devices may output stored and/or transmitted information as sounds or images. As electronic devices are highly integrated, and high-speed, high-volume wireless communication becomes commonplace, an electronic device, such as a mobile communication terminal, is recently being equipped with various functions. For example, an electronic device comes with the integrated functionality, including an entertainment function, such as playing video games, a multimedia function, such as replaying music/videos, a communication and security function for mobile banking, and a scheduling or e-wallet function. Such electronic devices become compact enough for users to carry in a convenient way.

(4) As recent trends are toward compact, slim, and high-density electronic products, metal frames or elements included in electronic products are made compact and slim. Depending on the characteristics of metal, the metal frame or elements may be formed of various types of metallic materials.

(5) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

DISCLOSURE

Technical Problem

- (6) An electronic device may provide various functions to the user through wireless communication. The electronic device may include antennas of various shapes or structures for communication through various frequencies. According to the conventional technology, there may be many difficulties in securing a space for installing an antenna in an electronic device, e.g., a mobile communication terminal. For example, due to its reduced size, it may be difficult to secure a sufficient physical spacing between circuit devices and antennas. Further, if a radiator included as at least a portion of the antenna is placed close to the surface of the electronic device, it is possible to secure a physical spacing between the radiator included as at least part of the antenna and circuit devices (e.g., a printed circuit board (PCB) line, a PCB ground, a semiconductor chip, a resistor, a capacitor, or an inductor), but radiation performance may be distorted due to the human body.
- (7) A general electronic device has an external appearance, and the external appearance of the electronic device may be affected by the material or arrangement of components. As part of the exterior of the electronic device, a plurality of members related to the exterior may be included. For example, a display, a receiver, or a front camera may be disposed on the front surface of the exterior of the electronic device, and a rear camera, a flash, and a speaker hole may be disposed on the rear surface of the exterior. Further, a side key, a universal serial bus (USB) connector, an earphone jack connector, a microphone hole, and the like may be disposed on the side surface of the electronic device. The arrangement of the components may be changed according to the exterior design of the electronic product.
- (8) The components related to the exterior, disposed on the electronic device, may have exterior members. For example, among the exterior members, a movable exterior member performing a physical pressing action or an exterior member having a hole for insertion may have weak waterproofing. As another example, the volume control key or the power key disposed on the side surface of the electronic device may be movably disposed with the key top exposed to the outside. A gap may occur between the moving member and the case frame and may cause penetration of a foreign body, such as moisture, into the electronic device, damaging, e.g., the substrate.
- (9) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device having a superior waterproofing structure while securing antenna performance.
- (10) In the case of the conventional coupling, a metal housing (hereinafter, 'metal') forming the exterior of the electronic device and a shape for the coupling structure are required, so that, for waterproofing and rigidity, a segment formed of one injection molding material is required, and the antenna radiation range is limited, resulting in a space limit in design. The presence of metal may interfere with the radiation range of the antenna, and the glass supporting the antenna module, the metal and the segment may not provide sufficient waterproofing.
- (11) In the conventional way of equipping an antenna, the material surrounding the periphery may be formed into a single material to secure the reliability of the instrument structure, such as waterproofing performance. For example, if a single material with about 45% polybutylene terephthalate (PBT)/glass fiber (GF) and a dielectric constant of about 3.5 is used, a margin for securing antenna performance may be insufficient.
- (12) Through the disclosure, for waterproofing and rigidity, two injection molded members formed of different injection molding materials may be coupled to the area bonded to the metal, and the dielectric constants of the injection molded members may be different from each other. The material that is bonded to the metal at the inside supplements the mechanical rigidity, preventing cracks and reinforcing the rigidity. The material bonded to the metal at the outside may increase the dielectric constant, unlike the internal material, and thus help secure antenna performance.
- (13) Materials having different dielectric constants may be applied to the plurality of injection

molded members, and different raw materials may be applied thereto. When different raw materials are applied, chemical bonding may not be completely achieved during injection molding.

Therefore, a physical coupling structure is formed to secure mechanical rigidity.

(14) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

Technical Solution

(15) In accordance with an aspect of the disclosure, a portable electronic device is provided. The portable electronic device includes a metal housing, an antenna module positioned in the metal housing, a first injection-molded member positioned in a radiation direction of the antenna module and having a first dielectric constant, a second injection-molded member at least partially in contact with the first injection-molded member, positioned in the radiation direction of the antenna module, and having a second dielectric constant different from the first dielectric constant, and a first organic film disposed along an inner surface of the metal housing to bond with at least a portion of the first injection-molded member. The second injection-molded member may form, together with the metal housing, an exterior of the portable electronic device. The second injection-molded member may be coupled with the first injection-molded member through a coupling structure in which one of the first injection-molded member or the second injection-molded member is at least partially protruding into another one of the first injection-molded member or the second injection-molded member.

(16) In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a metal housing, an antenna module positioned in the metal housing, a first injection-molded member including an inorganic material and having a first dielectric constant, and a second injection-molded member having a second dielectric constant higher than the first dielectric constant. The second injection-molded member may form, together with the metal housing, an exterior of the portable electronic device. The second injection-molded member may be at least partially bonded with the first injection-molded member through a coupling structure with the first injection-molded member.

Advantageous Effects

(17) According to various embodiments of the disclosure, as the first injection-molded member and the second injection-molded member are positioned in the radiation direction of the antenna module, it is possible to secure antenna radiation performance.

(18) According to various embodiments of the disclosure, it is possible to provide high rigidity through a coupling structure between the first injection-molded member and the second injection-molded member having different dielectric constants.

(19) According to various embodiments of the disclosure, it is possible to provide high waterproofing performance due to the second injection-molded member present on the exterior of the portable electronic device.

(20) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

(1) DESCRIPTION OF THE DRAWINGS

(2) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

(3) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

- (4) FIG. 2 is a front perspective view illustrating an electronic device according to an embodiment of the disclosure;
- (5) FIG. 3 is a rear perspective view illustrating an electronic device according to an embodiment of the disclosure;
- (6) FIG. 4 is an exploded perspective view illustrating an electronic device according to an embodiment of the disclosure;
- (7) FIG. 5 illustrates an exterior of an antenna module according to an embodiment of the disclosure;
- (8) FIG. 6 illustrates a conventional structure including an antenna module according to an embodiment of the disclosure;
- (9) FIG. 7 illustrates an enhanced structure for securing antenna performance according to an embodiment of the disclosure;
- (10) FIGS. 8A, 8B, and 8C illustrate a coupling structure of an injection-molded member according to various embodiments of the disclosure;
- (11) FIGS. 9A, 9B, and 9C illustrate a coupling structure of an injection-molded member according to various embodiments of the disclosure;
- (12) FIG. 10 illustrates a manufacturing process using injection-molding materials with different dielectric constants according to an embodiment of the disclosure;
- (13) FIGS. 11A and 11B illustrate an antenna module stack structure where an additional component may be attached according to various embodiments of the disclosure; and
- (14) FIG. 12 illustrates a coupling structure of a first injection-molded member and a second injection-molded member, according to an embodiment of the disclosure.
- (15) Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

MODE FOR INVENTION

(16) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

(17) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(18) It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

(19) FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

(20) Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**,

an interface **177**, a connection terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one (e.g., the connection terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. According to an embodiment, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be integrated into a single component (e.g., the display module **160**).

(21) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be configured to use lower power than the main processor **121** or to be specified for a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(22) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. The artificial intelligence model may be generated via machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(23) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(24) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(25) The input module **150** may receive a command or data to be used by another component (e.g.,

the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, keys (e.g., buttons), or a digital pen (e.g., a stylus pen).

(26) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(27) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor configured to detect a touch, or a pressure sensor configured to measure the intensity of a force generated by the touch.

(28) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(29) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(30) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(31) A connection terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connection terminal **178** may include, for example, a HDMI connector, a USB connector, an SD card connector, or an audio connector (e.g., a headphone connector).

(32) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(33) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(34) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(35) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(36) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101**

and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via a first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5th generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and/or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(37) The wireless communication module **192** may support a 5G network, after a 4th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(38) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module may include an antenna including a radiator formed of a conductor or conductive pattern formed on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., an antenna array). In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected from the plurality of antennas by, e.g., the communication module **190**. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **197**.

(39) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit

board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(40) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(41) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. The external electronic devices **102** or **104** each may be a device of the same or a different type from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra-low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or health-care) based on 5G communication technology or IoT-related technology.

(42) The electronic device according to various embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(43) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it

means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(44) As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(45) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(46) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(47) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(48) FIG. 2 is a front perspective view illustrating an electronic device according to an embodiment of the disclosure. FIG. 3 is a rear perspective view illustrating an electronic device according to an embodiment of the disclosure.

(49) Referring to FIGS. 2 and 3, according to an embodiment, an electronic device **101** may include a housing **310** with a front surface **310A**, a rear surface **310B**, and a side surface **310C** surrounding a space between the front surface **310A** and the rear surface **310B**. According to another embodiment (not shown), the housing **310** may denote a structure forming part of the front surface **310A**, the rear surface **310B**, and the side surface **310C** of FIG. 2. According to an embodiment, at least part of the front surface **310A** may have a substantially transparent front plate

302 (e.g., a glass plate or polymer plate including various coat layers). The rear surface **310B** may be formed by a rear plate **311**. The rear plate **311** may be formed of, e.g., glass, ceramic, polymer, metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two thereof. The side surface **310C** may be formed by a side bezel structure (or a “side member”) **318** that couples to the front plate **302** and the rear plate **311** and includes a metal and/or polymer. According to an embodiment, the rear plate **311** and the side bezel structure **318** may be integrally formed together and include the same material (e.g., glass, metal, such as aluminum, or ceramic).

(50) In the embodiment illustrated, the front plate **302** may include two first edge areas **310D**, which seamlessly and bendingly extend from the front surface **310A** to the rear plate **311**, on both the long edges of the front plate **302**. In the embodiment (refer to FIG. 3) illustrated, the rear plate **311** may include two second edge areas **310E**, which seamlessly and bendingly extend from the rear surface **310B** to the front plate, on both the long edges. According to an embodiment, the front plate **302** (or the rear plate **311**) may include only one of the first edge areas **310D** (or the second edge areas **310E**). Alternatively, the first edge areas **310D** or the second edge areas **310E** may partially be excluded. According to an embodiment, at side view of the electronic device **101**, the side bezel structure **318** may have a first thickness (or width) for sides that do not have the first edge areas **310D** or the second edge areas **310E** and a second thickness, which is smaller than the first thickness, for sides that have the first edge areas **310D** or the second edge areas **310E**.

(51) According to an embodiment, the electronic device **101** may include at least one of a display **301** (e.g., the display module **160** of FIG. 1), audio modules **303**, **307**, and **314** (e.g., the audio module **170** of FIG. 1), a sensor module (e.g., the sensor module of FIG. 1), **176**), camera modules **305**, **312**, and **313** (e.g., the camera module **180** of FIG. 1), a key input device **317** (e.g., the input module **150** of FIG. 1), and connector holes **308** and **309** (e.g., the connection terminal **178** of FIG. 1). According to an embodiment, the electronic device **101** may exclude at least one (e.g., the connector hole **309**) of the components or may add other components.

(52) According to an embodiment, the display **301** may be visually exposed through, e.g., a majority portion of the front plate **302**. According to an embodiment, at least a portion of the display **301** may be exposed through the front plate **302** forming the front surface **310A** and the first edge areas **310D**. According to an embodiment, the edge of the display **301** may be formed to be substantially the same in shape as an adjacent outer edge of the front plate **302**. According to another embodiment (not shown), the interval between the outer edge of the display **301** and the outer edge of the front plate **302** may remain substantially even to give a larger area of exposure the display **301**.

(53) According to an embodiment, the surface (or the front plate **302**) of the housing **310** may include a screen display area formed as the display **301** is visually exposed. For example, the screen display area may include the front surface **310A** and first edge areas **310D**.

(54) According to an embodiment, a recess or opening may be formed in a portion of the screen display area (e.g., the front surface **310A** or the first edge area **310D**) of the display **301**, and at least one or more of the audio module **314**, sensor module (not shown), light emitting device (not shown), and camera module **305** may be aligned with the recess or opening. According to another embodiment (not shown), at least one or more of the audio module **314**, sensor module (not shown), camera module **305**, fingerprint sensor (not shown), and light emitting device (not shown) may be included on the rear surface of the screen display area of the display **301**.

(55) According to an embodiment, at least one camera module **305** and **312** may be disposed under the display **301**. For example, the first camera module **305** may be disposed in at least a partial area of the display **301** corresponding to a camera field of view (FOV). As the first camera module **305** is disposed in at least a partial area of the display **301** corresponding to the camera field of view (FOV), the position of the first camera module **305** may not be visually distinguished (or exposed). According to an embodiment, when the display **301** is viewed from the front surface **310A**, the first camera module **305** is disposed in a portion corresponding to the camera field of view (FOV),

which is at least a portion of the display **301**, so that the first camera module **305** may obtain an image of the external subject while not being visually exposed to the outside. For example, the first camera module **305** may be an under display camera (UDC).

(56) In an embodiment, the electronic device **101** may include a display (not shown) that is slidably disposed and provides a screen (e.g., a display area). For example, the display area of the electronic device **101** may be an area that is visually exposed and enables an image to be output. In one example, the electronic device **101** may adjust the display area according to the movement of a sliding plate (not shown) or the movement of the display. For example, the electronic device **101** may include a rollable electronic device configured to selectively expand the display area as at least a portion (e.g., the housing **310**) of the electronic device **101** is operated to be at least partially slidable. The above-described display **301** may be referred to as, e.g., a slide-out display or an expandable display.

(57) According to another embodiment (not shown), the display **301** may be disposed to be coupled with, or adjacent, a touch detecting circuit, a pressure sensor capable of measuring the strength (pressure) of touches, and/or a digitizer for detecting a magnetic field-type stylus pen. According to an embodiment, at least part of the key input device **317** may be disposed in the first edge areas **310D** and/or the second edge areas **310E**.

(58) According to an embodiment, the audio modules **303**, **307**, and **314** may include, e.g., a microphone hole **303** and speaker holes **307** and **314**. The microphone hole **303** may have a microphone inside to obtain external sounds. According to an embodiment, there may be a plurality of microphones to be able to detect the direction of a sound. The speaker holes **307** and **314** may include an external speaker hole **307** and a phone receiver hole **314**. In some embodiments, the speaker holes **307** and **314** and the microphone hole **303** may be implemented as a single hole, or a speaker may be included without the speaker holes **307** and **314** (e.g., a piezo speaker). The audio modules **303**, **307**, and **314** are not limited to the above-described structure. Depending on the structure of the electronic device **101**, various design changes may be made—e.g., only some of the audio modules may be mounted, or a new audio module may be added.

(59) According to an embodiment, the sensor modules (not shown) may generate an electrical signal or data value corresponding to an internal operating state or external environmental state of the electronic device **101**. The sensor modules (not shown) may include a first sensor module (e.g., a proximity sensor) and/or a second sensor module (e.g., a fingerprint sensor) disposed on the front surface **310A** of the housing **310** and/or a third sensor module (e.g., a heart rate monitor (HRM) sensor) and/or a fourth sensor module (e.g., a fingerprint sensor) disposed on the rear surface **310B** of the housing **310**. In an embodiment (not shown), the fingerprint sensor may be disposed on the rear surface **310B** as well as on the front surface **310A** (e.g., the display **301**) of the housing **310**. The electronic device **101** may include a sensor module not shown, e.g., at least one of a gesture sensor, a gyro sensor, a barometric sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor. The sensor modules are not limited to the above-described structure. Depending on the structure of the electronic device **101**, various design changes may be made—e.g., only some of the sensor modules may be mounted, or a new sensor module may be added.

(60) According to an embodiment, the camera modules **305**, **312**, and **313** may include a first camera module **305** disposed on the front surface **310A** of the electronic device **101**, and a rear camera device **312** and/or a flash **313** disposed on the rear surface **310B**. The camera modules **305** and **312** may include one or more lenses, an image sensor, and/or an image signal processor. The flash **313** may include, e.g., a light emitting diode (LED) or a xenon lamp. According to an embodiment, two or more lenses (an infrared (IR) camera, a wide-angle lens, and a telescopic lens) and image sensors may be disposed on one surface of the electronic device **101**. The camera modules **305**, **312**, and **313** are not limited to the above-described structure. Depending on the structure of the electronic device **101**, various design changes may be made—e.g., only some of the

camera modules may be mounted, or a new camera module may be added.

(61) According to an embodiment, the electronic device **101** may include a plurality of camera modules (e.g., a dual camera or triple camera) having different attributes (e.g., angle of view) or functions. For example, a plurality of camera modules **305** and **312** including lenses having different angles of view may be configured, and the electronic device **101** may control to change the angle of view of the camera modules **305** and **312** performed by the electronic device **101** based on the user's selection. For example, at least one of the plurality of camera modules **305** and **312** may be a wide-angle camera and at least another of the plurality of camera modules may be a telephoto camera. Similarly, at least one of the plurality of camera modules **305** and **312** may be a front camera and at least another of the plurality of camera modules may be a rear camera. Further, the plurality of camera modules **305** and **312** may include at least one of a wide-angle camera, a telephoto camera, and an infrared (IR) camera (e.g., a time of flight (TOF) camera, a structured light camera). According to an embodiment, the IR camera may be operated as at least a portion of the sensor module. For example, the TOF camera may be operated as at least a portion of a sensor module (not shown) for detecting the distance to the subject.

(62) According to an embodiment, the key input device **317** may be disposed on the side surface **310C** of the housing **310**. According to an embodiment, the electronic device **101** may exclude all or some of the above-mentioned key input devices **317** and the excluded key input devices **317** may be implemented in other forms, e.g., as soft keys, on the display **301**. According to an embodiment, the key input device may include the sensor module (not shown) disposed on the rear surface **310B** of the housing **310**.

(63) According to an embodiment, the light emitting device (not shown) may be disposed on, e.g., the front surface **310A** of the housing **310**. The light emitting device (not shown) may provide, e.g., information about the state of the electronic device **101** in the form of light. According to another embodiment, the light emitting device (not shown) may provide a light source that interacts with, e.g., the front camera module **305**. The light emitting device (not shown) may include, e.g., a light emitting device (LED), an infrared (IR) LED, and/or a xenon lamp.

(64) According to an embodiment, the connector holes **308** and **309** may include, e.g., a first connector hole **308** for receiving a connector (e.g., a universal serial bus (USB) connector) for transmitting or receiving power and/or data to/from an external electronic device and/or a second connector hole (e.g., an earphone jack) **309** for receiving a connector for transmitting or receiving audio signals to/from the external electronic device.

(65) According to an embodiment, some of the camera modules **305** and **312** and/or some of the sensor modules (not shown) may be disposed to be exposed to the outside through at least a portion of the display **301**. For example, the camera module **305** may include a punch hole camera disposed inside a hole or recess formed in the rear surface of the display **301**. According to an embodiment, the camera module **312** may be disposed inside the housing **310** so that the lens is exposed to the rear surface **310B** of the electronic device **101**. For example, the camera module **312** may be disposed on a printed circuit board (e.g., the printed circuit board **340** of FIG. 4).

(66) According to an embodiment, the camera module **305** and/or the sensor module may be disposed to contact with the external environment through a transparent area from the internal space of the electronic device **101** to the front plate **302** of the display **301**. Further, some sensor module (not shown) may be disposed to perform its functions without being visually exposed through the front plate **302** in the internal space of the electronic device.

(67) FIG. 4 is an exploded perspective view illustrating an electronic device according to an embodiment of the disclosure.

(68) Referring to FIG. 4, according to various embodiments, an electronic device **101** (e.g., the electronic device **101** of FIGS. 1 to 3) may include a side bezel structure **331** (e.g., the side bezel structure **318** of FIG. 2), a first supporting member **372**, a front plate **320** (e.g., the front plate **302** of FIG. 2), a display **330** (e.g., the display **301** of FIG. 2), a printed circuit board **340** (e.g., a PCB,

flexible PCB (FPCB), or rigid flexible PCB (RFPCB)), a battery **350** (e.g., the battery **189** in FIG. **1**), a second supporting member **360** (e.g., a rear case), an antenna **370** (e.g., the antenna module **197** of FIG. **1**), and a rear plate **380** (e.g., the rear plate **311** of FIG. **2**). According to an embodiment, the electronic device **101** may exclude at least one (e.g., the first supporting member **372** or second supporting member **360**) of the components or may add other components. At least one of the components of the electronic device **101** may be the same or similar to at least one of the components of the electronic device **101** of FIG. **2** or **3** and no duplicate description is made below.

(69) According to various embodiments, the first supporting member **372** may be disposed inside the electronic device **101** to be connected with the side bezel structure **331** or integrated with the side bezel structure **331**. The first supporting member **372** may be formed of, e.g., a metal and/or non-metallic material (e.g., polymer). The display **330** may be joined onto one surface of the first supporting member **372**, and the printed circuit board **340** may be joined onto the opposite surface of the first supporting member **372**.

(70) According to various embodiments, a processor, a memory, and/or an interface may be mounted on the printed circuit board **340**. The processor may include one or more of, e.g., a central processing unit, an application processor, a graphic processing device, an image signal processing, a sensor hub processor, or a communication processor. According to various embodiments, the printed circuit board **340** may include a flexible printed circuit board type radio frequency cable (FRC). For example, the printed circuit board **340** may be disposed on at least a portion of the first supporting member **372** and may be electrically connected with an antenna module (e.g., the antenna module **197** of FIG. **1**) and a communication module (e.g., the communication module **190** of FIG. **1**).

(71) According to an embodiment, the memory (e.g., the memory **130** of FIG. **1**) may include, e.g., a volatile or non-volatile memory.

(72) According to an embodiment, the interface (e.g., the interface **177** of FIG. **1**) may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, and/or an audio interface. The interface may electrically or physically connect, e.g., the electronic device **101** with an external electronic device and may include a USB connector, an SD card/multimedia card (MMC) connector, or an audio connector.

(73) According to various embodiments, the battery **350** (e.g., the battery **189** of FIG. **1**) may be a device for supplying power to at least one component of the electronic device **101**. The battery **189** may include, e.g., a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell. At least a portion of the battery **350** may be disposed on substantially the same plane as the printed circuit board **340**. The battery **350** may be integrally or detachably disposed inside the electronic device **101**.

(74) According to various embodiments, the second supporting member **360** (e.g., a rear case) may be disposed between the printed circuit board **340** and the antenna **370**. For example, the second supporting member **360** may include one surface to which at least one of the printed circuit board **340** and the battery **350** is coupled, and another surface to which the antenna **370** is coupled.

(75) According to various embodiments, the antenna **370** (e.g., the antenna module **197** of FIG. **1**) may be disposed between the rear plate **380** and the battery **350**. The antenna **370** may include, e.g., a near-field communication (NFC) antenna, a wireless charging antenna, and/or a magnetic secure transmission (MST) antenna. The antenna **370** may perform short-range communication with, e.g., an external device or may wirelessly transmit or receive power necessary for charging. According to an embodiment, an antenna structure may be formed by a portion or combination of the side bezel structure **331** and/or the first supporting member **372**.

(76) According to various embodiments, the rear plate **380** may form at least a portion of the rear surface (e.g., the rear surface **310B** of FIG. **3**) of the electronic device **101**.

(77) FIG. **5** illustrates an exterior of an antenna module according to an embodiment of the disclosure.

(78) Referring to FIG. 5, a metal **501** and a module stack structure **502** forming the exterior of the electronic device **101** are illustrated. The disclosure proposes an instrument structure optimized for enhancing antenna performance and securing waterproofing performance.

(79) FIG. 6 illustrates a conventional structure including an antenna module according to an embodiment of the disclosure.

(80) FIG. 7 illustrates an enhanced structure for securing antenna performance according to an embodiment of the disclosure.

(81) Referring to FIG. 6, the conventional structure of the module stack structure **502** includes an antenna module (e.g., **601** of FIG. 6), a lower glass (e.g., the back glass **602** of FIG. 6) positioned in an area including the radiation direction, under the antenna module (e.g., **601** of FIG. 6), a metal (e.g., **604** of FIG. 6) forming the exterior of the electronic device **101**, and an injection-molded member (e.g., **603** of FIG. 6) positioned between the antenna module (e.g., **601** of FIG. 6) and the metal (e.g., **604** of FIG. 6).

(82) According to various embodiments, the injection-molded member (e.g., **603** of FIG. 6) included in the conventional structure contains glass fiber as a thermoplastic resin. The glass fiber plays a role to increase the rigidity of the material to secure the strength of the device and to reduce shrinkage to secure durability and set deformation and/or prevent set damage to thereby secure waterproofing performance. Further, the material surrounding the periphery of the module stack structure **502**, which is the antenna mounting structure, is formed into a single material to secure reliability of the instrument structure, such as waterproofing performance.

(83) According to various embodiments, to increase the performance of the antenna, it is necessary to secure an antenna performance influence section (e.g., **610** of FIG. 6). Due to the conductive metal (e.g., **604** in FIG. 6), the antenna performance influence section (e.g., **610** of FIG. 6) may be partially limited. If a broader area is secured as the radiation area using a non-conductive injection-molded material, the performance of the antenna may be enhanced.

(84) According to various embodiments, the performance of the antenna may be related to the dielectric constant of the injection-molded member (e.g., **603** of FIG. 6). As a raw material (e.g., high molecular polymer or resin) of the injection-molded member (e.g., **603** of FIG. 6), a material having a dielectric constant of about 2.0 to 3.5 may be used, and it may contain an inorganic material. The glass fiber has a dielectric constant between about 7.2 and 7.3. Thus, if the proportion of the glass fiber is increased, the dielectric constant increases. In the case of a resin having a dielectric constant of about 3.8, a reference cumulative distribution function (CDF) value in the carrier's frequency band (e.g., n261) is 18.4 dbm, and it is experimentally identified that it is increased by about 0.48 dbm, thereby enhancing the antenna performance. The CDF may be a 5G antenna performance evaluation measurement scheme. In general, a measurement value of about 50% may be used as a reference for determining the performance of the terminal. In other words, CDF 50% means that the performance of all 5G antenna beams is measured and, under the assumption that the total number of beams is 100, the antenna performance may be simulated for each dielectric constant through the equivalent isotropic radiated power (EIRP), which is the output power value of one beam with the 50th best performance. As an example, the gain for the antenna performance may be derived by performing simulation per dielectric constant for each of a plurality of frequency bands (e.g., n258 and n261). As the dielectric constant is gradually increased, it may be identified that the sum of the gain values is increased, which may be advantageous in securing performance. When the dielectric constant of the injection-molded member (e.g., **603** of FIG. 6) is increased, the antenna performance may be enhanced. However, the fluidity of the resin is decreased, so that it is required to increase the temperature of injection molding. The increase in temperature may cause carbonization and poor appearance. Further, to enhance the fluidity of the resin and prevent injection molding at high temperature, a ceramic-based inorganic material, such as zirconia and alumina, having a particle size smaller than that of glass fiber may be added. If the inorganic material is added, fluidity may be better than that of glass fiber, but the ceramic-based

inorganic material has a problem of having lower impact resistance and rigidity than that of glass fiber. If the rigidity is low, a crack may occur due to a fall, and it may be difficult to secure waterproofing performance. An enhanced structure may be described with reference to FIG. 7 so as to reinforce rigidity and waterproofing characteristics while securing the performance of the antenna.

(85) Referring to FIG. 7, an enhanced structure inside the module stack structure **502** includes an antenna module (e.g., **601** of FIG. 6), a lower glass (e.g., the back glass **602** of FIG. 6) positioned in an area under the antenna module (e.g., **601** of FIG. 6), a metal housing (e.g., **704** of FIG. 7) forming the exterior of the electronic device **101**, and a first injection-molded member (e.g., **701** of FIG. 7) positioned between the antenna module (e.g., **601** of FIG. 6) and the metal housing (e.g., **704** of FIG. 7). The electronic device **101** includes the metal housing (e.g., **704** of FIG. 7) and a second injection-molded member (e.g., **702** of FIG. 7) at the periphery of the electronic device **101**.

(86) According to various embodiments, the description of FIG. 6 may be applied to the antenna module (e.g., **601** of FIG. 6) and the lower glass (e.g., the back glass **602** of FIG. 6). In FIG. 7, the antenna module (e.g., **601** of FIG. 6) may be spaced, a predetermined interval, apart from the lower glass (e.g., the back glass **602**), as compared with the conventional structure (e.g., FIG. 6), so as to secure a radiation area (e.g., **710** of FIG. 7). For the antenna module (e.g., **601** of FIG. 6), the first injection-molded member (e.g., **701** of FIG. 7) and the second injection-molded member (e.g., **702** of FIG. 7) placed on the radiation area (e.g., **710** of FIG. 7) may be said to be mounted in the x-axis direction. For each component, the injection-molded members may be mounted with respect to the x-axis direction, such as the antenna module (e.g., **601** of FIG. 6), to minimize the loss of waterproofing and antenna radiation performance, and the area of the conductive metal housing (e.g., **704** of FIG. 7) may be minimized.

(87) According to various embodiments, the metal housing (e.g., **704** of FIG. 7) may be bonded to the two injection-molded members. The two injection-molded members may have different dielectric constants. The first injection-molded member (e.g., **701** of FIG. 7) bonded inside the metal housing (e.g., **704** of FIG. 7) may supplement the mechanical rigidity, preventing cracks and reinforcing rigidity. The first injection-molded member (e.g., **701** of FIG. 7) bonded inside the metal housing (e.g., **704** of FIG. 7) may form a first organic film (e.g., **703** of FIG. 7). The second injection-molded member (e.g., **702** of FIG. 7) that, together with the metal housing (e.g., **704** of FIG. 7), forms the exterior of the electronic device **101** and is bonded to the metal housing (e.g., **704** of FIG. 7) may play a role to increase the dielectric constant over the first injection-molded member (e.g., **701** of FIG. 7) to secure antenna performance. The second injection-molded member (e.g., **702** of FIG. 7) bonded to the metal housing (e.g., **704** of FIG. 7) may form a second organic film (not shown) on at least a portion of the bonding surface. The first organic film (e.g., **704** of FIG. 7) may be formed through a TRI process, and the second organic film (not shown) may be formed after the first organic film (e.g., **704** of FIG. 7) is formed. The first organic film or the second organic film included in the metal housing (e.g., **704** of FIG. 7) may enhance bonding strength between the metal housing (e.g., **704** of FIG. 7) and the injection-molded member.

(88) According to various embodiments, the metal housing (e.g., **704** of FIG. 7) may be formed symmetrically with respect to the x-axis to not interfere with the radiation area (e.g., **710** of FIG. 7). It may be seen that the radiation area (e.g., **710** of FIG. 7) has a wider section than the antenna performance influence section (e.g., **610** of FIG. 6) in the conventional structure (e.g., FIG. 6).

(89) According to various embodiments, the description of the injection-molded member (e.g., **603** of FIG. 6) of FIG. 6 may be applied to the first injection-molded member (e.g., **701** of FIG. 7). The first injection-molded member may be mounted inside the metal housing (e.g., **704** of FIG. 7) and may form the metal housing (e.g., **704** of FIG. 7) and the first organic film (e.g., **703** of FIG. 7). The first injection-molded member (e.g., **701** of FIG. 7) may include a thermoplastic resin and an inorganic material (e.g., primary PBT, secondary PC). The inorganic material may increase the rigidity of the injection-molded member to secure the rigidity of the device and may reduce

shrinkage to reduce set deformation. Further, the inorganic material may secure the durability of the device and prevent damage to the device to secure waterproofing performance. If the content of the inorganic material is increased, the fluidity of the injection-molded product may decrease, so that difficulty may occur in injection molding due to insufficient fluidity. If the temperature is raised to increase fluidity, carbonization and poor appearance may occur. To secure the rigidity of the instrument of the module stack structure **502**, waterproofing performance, and antenna performance, the first injection-molded member (e.g., **701** of FIG. 7) present therein may include a material having a predetermined degree of rigidity or more using an inorganic material, and the first injection-molded member (e.g., **701** of FIG. 7) may be bonded to the metal housing (e.g., **704** of FIG. 7) and implement mechanical rigidity while implementing waterproofing performance. The resin applied to the first injection-molded member (e.g., **701** of FIG. 7) may contain about 30% or more of an inorganic material, such as glass fiber, to secure mechanical rigidity and prevent shrinkage. The first injection-molded member (e.g., **701** of FIG. 7) may form a coupling structure (or an interlocked structure) for coupling with the second injection-molded member (e.g., **702** of FIG. 7) having a different dielectric constant.

(90) According to various embodiments, the second injection-molded member (e.g., **702** of FIG. 7) may use a material having a higher dielectric constant than the first injection-molded member (e.g., **701** of FIG. 7) to enhance antenna performance. The second injection-molded member (e.g., **702** of FIG. 7), as a component for securing antenna performance, may have a composition different from the material used for the first injection-molded member (e.g., **701** of FIG. 7). The resin raw material base of the second injection-molded member (e.g., **702** of FIG. 7) may include, e.g., polycarbonate, poly butylene terephthalate, or poly phenylene sulfide. The inorganic material may also include, e.g., glass fiber, talc, wollastonite, and ZrO₂. The second injection-molded member (e.g., **702** of FIG. 7) is applied differently in at least one of the components included in the inorganic material and raw material base or be different in composition from the first injection-molded member (e.g., **701** in FIG. 7).

(91) According to various embodiments, the basic dielectric constant of each material may be as follows. As examples that may be used as raw materials, polycarbonate may have a dielectric constant of about 2.9 to 3.0, polypropylene may have a dielectric constant of about 2.1 to 2.2, modified poly phenylene oxide may have a dielectric constant of about 2.0 to 2.1, poly butylene terephthalate may have a dielectric constant of about 3.0 to 3.2, and poly phenylene sulfide may have a dielectric constant of about 3.1 to 3.3. As example that may be the inorganic material, glass fiber may have a dielectric constant of about 7.2 to 7.3, minerals, such as talc/wollastonite, may have a dielectric constant of about 4.5 to 6.5, and ZrO₂ (ceramic-based materials, such as zirconia) may have a dielectric constant of about 4.6 to 5.0. There is a difference in intrinsic dielectric constant of each raw material and inorganic material. Based on this difference, a material is made with a different composition from the first injection-molded member (e.g., **701** of FIG. 7), so as to utilize the second injection-molded member (e.g., **702** of FIG. 7) having a different dielectric constant from the first injection-molded member (e.g., **701** of FIG. 7), and the antenna characteristics may be optimized by increasing the dielectric constant of the second injection-molded member (e.g., **702** of FIG. 7).

(92) According to various embodiments, since injection-molded members having different dielectric constants are difficult to completely chemically couple during injection molding, it is possible to secure mechanical rigidity by forming a physical coupling structure (or interlocked structure).

(93) FIGS. **8A**, **8B**, and **8C** illustrate a coupling structure of an injection-molded member according to various embodiments of the disclosure.

(94) FIGS. **9A**, **9B**, and **9C** illustrate a coupling structure of an injection-molded member according to various embodiments of the disclosure.

(95) FIG. **10** illustrates a manufacturing process using injection-molding materials with different

dielectric constants according to an embodiment of the disclosure.

(96) FIG. 12 illustrates a coupling structure of a first injection-molded member and a second injection-molded member, according to an embodiment of the disclosure.

(97) According to various embodiments, the structure of the module stack structure 502 is illustrated at the outside of the electronic device 101. FIG. 8A illustrates a metal housing and a second injection-molded member (e.g., 702 of FIGS. 7 and 802 of FIG. 8A) exposed to the outside of the electronic device 101 and illustrates a first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) positioned inside the electronic device 101. The description of FIGS. 5 to 7 may be applied to the first injection-molded member (e.g., 801 of FIG. 8A) and the second injection-molded member (e.g., 802 of FIG. 8A). FIG. 8B is a cross-sectional view taken along AA' of FIG. 8A, illustrating a coupling structure (e.g., 810 of FIG. 8A) of the first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) and a second injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A). FIG. 8C illustrates a cross-section of the finished product.

(98) According to various embodiments, the coupling structure (e.g., 810 of FIG. 8B) is applied to the coupling of two injection-molded members having different dielectric constants and different raw material bases to ensure the durability of the device. It is shown that the coupling structure may be formed in various forms (e.g., 901a, 901b, and 901c) in FIGS. 9A to 9C. FIG. 8C is a cross-sectional view of the finished product 811, which may include various types of coupling structures (e.g., 810 of FIG. 8B, 901a of FIG. 9A, 901b of FIG. 9B, or 901c of FIG. 9C) mentioned above, between the plurality of injection-molded members. For example, the coupling structure (e.g., 810 of FIG. 8A) may be formed so that the first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) protrudes into the second injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A) so that the second injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A) surrounds the first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A). For example, the coupling structure (e.g., 810 of FIG. 8A) may be formed so that the second injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A) protrudes into the first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) so that the first injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) surrounds the second injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A). As an example, the coupling structure (e.g., 810 of FIG. 8B) may include a shape in which the second injection-molded member (e.g., 701 of FIG. 7 or 801 of FIG. 8A) penetrates the first injection-molded member (e.g., 702 of FIG. 7 or 802 of FIG. 8A). Various interlocked structures or coupling structures may be formed to supplement durability of the electronic device 101. If two injection-molded members with different dielectric constants are coupled in a face-to-face structure rather than in an interlocked structure, it may be observed with an optical microscope that micro gaps may occur between the injection-molded members, and if the instrument is dropped or tested for reliability, they may be separated at the bonding interface.

(99) According to various embodiments, a process of manufacturing a module stack structure 502 formed of two injection-molded members having different dielectric constants may be described with reference to FIG. 10. The manufacturing process of FIG. 10 is described. First, a first injection-molded member is inserted into a metal housing (e.g., 604 of FIG. 6) (e.g., 1001 of FIG. 10), a mounting space where a second injection-molded member (e.g., 702 of FIG. 7) may be positioned is secured through CNC processing (e.g., 1002 of FIG. 10), the second injection-molded member (e.g., 702 of FIG. 7) with a different dielectric constant is inserted (e.g., 1003 of FIG. 10), and a shape may be formed by CNC processing to secure a mounting space where an antenna module (e.g., 601 of FIG. 6) may be positioned (e.g., 1004 of FIG. 10).

(100) FIGS. 11A and 11B illustrate an antenna module stack structure where an additional component may be attached according to various embodiments of the disclosure.

(101) According to various embodiments, the description of FIGS. 5 to 7 may be applied to the metal housing (e.g., 604 of FIG. 6), the first injection-molded member (e.g., 701 of FIG. 7), and the second injection-molded member (e.g., 702 of FIG. 7) included in the metal (e.g., 501 of FIG. 5)

and the module stack structure (e.g., **502** of FIG. **5**).

(102) According to various embodiments, a film (e.g., **1101** of FIG. **11A**) may be positioned outside the electronic device **101** for waterproofing, rigidity, or decorative purposes without forming a separate coupling structure with components inside the module stack structure (e.g., **502** of FIG. **5**). The film may be attached onto the second injection-molded member (e.g., **702** of FIG. **7**) exposed to the outside of the electronic device **101** through, e.g., painting/deposition. The film (e.g., **1101** of FIG. **11A**) may implement various textures based on a metallic paint or vapor deposition. The film may include a plastic material to implement a metallic texture. The material (e.g., **1102** of FIG. **11B**) and/or tape for bonding may be disposed, for bonding, between the first injection-molded member (e.g., **701** of FIG. **7**) and the second injection-molded member (e.g., **702** of FIG. **7**). The film (e.g., **1101** of FIG. **11A**) is an example of the disclosure and may be formed of at least one of deposition and coating layers.

(103) According to various embodiments, a plurality of injection-molded members having different dielectric constants may form the coupling structure as shown in FIG. **12**. According to an embodiment, to form the module stack structure **502**, a plurality of injection-molded members may be included. For example, a first injection-molded member (e.g., **1201** of FIG. **12**), a second injection-molded member (e.g., **1202** of FIG. **12**), and a third injection-molded member (e.g., **1203** of FIG. **12**) may be coupled together. According to the disclosure, the first injection-molded member (e.g., **701** of FIG. **7**) may be formed to surround the second injection-molded member (e.g., **702** of FIG. **7**). Referring to FIG. **12**, the border between the first injection-molded member and the second injection-molded member (e.g., **1201** and **1202** of FIGS. **12** or **1202** and **1203** of FIG. **12**) may form a coupling structure in which the two injection-molded members with different dielectric constants are interlocked for durability.

(104) According to various embodiments of the disclosure, there may be proposed a portable electronic device comprising a metal housing, an antenna module positioned in the metal housing, a first injection-molded member positioned in a radiation direction of the antenna module and having a first dielectric constant, a second injection-molded member at least partially in contact with the first injection-molded member, positioned in the radiation direction of the antenna module, and having a second dielectric constant different from the first dielectric constant, and a first organic film disposed along an inner surface of the metal housing to bond with at least a portion of the first injection-molded member, wherein the second injection-molded member forms, together with the metal housing, an exterior of the portable electronic device, and wherein the second injection-molded member is coupled with the first injection-molded member through a coupling structure in which one of the first injection-molded member or the second injection-molded member is at least partially protruding into another one of the first injection-molded member or the second injection-molded member.

(105) According to various embodiments, the portable electronic device may further comprise a second organic film disposed along the inner surface of the metal housing to bond with at least a portion of the second injection-molded member.

(106) According to various embodiments, the metal housing may be positioned on an upper end and a lower end of the second injection-molded member with respect to the radiation direction of the antenna module.

(107) According to various embodiments, a composition ratio for a raw material and an inorganic material included in the first injection-molded member may differ from that of the second injection-molded member, and types for a raw material and an inorganic material included in the first injection-molded member may differ from those included in the second injection-molded member, and the first dielectric constant may differ from the second dielectric constant by a dielectric constant difference of 0.5 or more. The second injection-molded member may have a higher dielectric constant than the first injection-molded member.

(108) According to various embodiments, in the coupling structure, the second injection-molded

member may protrude into the first injection-molded member.

(109) According to various embodiments, in the coupling structure, the first injection-molded member may protrude into the second injection-molded member.

(110) According to various embodiments, the portable electronic device may further comprise at least one of a film, a deposited layer, or a painted layer surrounding at least a portion of a surface of the second injection-molded member forming the exterior of the portable electronic device.

(111) According to various embodiments, the coupling structure may include a shape in which the one of the first injection-molded member or the second injection-molded member penetrates into the other one of the first injection-molded member or the second injection-molded member.

(112) According to various embodiments, a portable electronic device comprises a metal housing, an antenna module positioned in the metal housing, a first injection-molded member including an inorganic material and having a first dielectric constant, and a second injection-molded member having a second dielectric constant higher than the first dielectric constant, wherein the second injection-molded member may form, together with the metal housing, an exterior of the portable electronic device, and wherein the second injection-molded member be at least partially bonded with the first injection-molded member through a coupling structure with the first injection-molded member.

(113) According to various embodiments, the portable electronic device may further comprise a first organic film disposed along an inner surface of the metal housing to bond with at least a portion of the first injection-molded member.

(114) According to various embodiments, the portable electronic device may further comprise a second organic film disposed along the inner surface of the metal housing to bond with at least a portion of the second injection-molded member.

(115) According to various embodiments, the first injection-molded member and the second injection-molded member may be positioned in a radiation direction of the antenna module.

(116) According to various embodiments, a composition ratio for a raw material and an inorganic material included in the first injection-molded member may differ from that of the second injection-molded member.

(117) According to various embodiments, types for a raw material and an inorganic material included in the first injection-molded member may differ from those included in the second injection-molded member.

(118) According to various embodiments, in the coupling structure, the second injection-molded member may protrude into the first injection-molded member.

(119) According to various embodiments, in the coupling structure, the first injection-molded member may protrude into the second injection-molded member.

(120) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. A portable electronic device comprising: a metal housing; antenna circuitry positioned in the metal housing; a first injection-molded member positioned in a radiation direction of the antenna circuitry and having a first dielectric constant; a second injection-molded member at least partially in contact with the first injection-molded member, positioned in the radiation direction of the antenna circuitry, and having a second dielectric constant greater than the first dielectric constant; a first organic film disposed along and contacting at least one first part of an inner surface of the metal housing to bond with at least a portion of the first injection-molded member; and a second organic film disposed along and contacting at least one second part of the inner surface of the metal

housing to bond with at least a portion of the second injection-molded member, wherein the first injection-molded member, the second injection-molded member, and the antenna circuitry are disposed along a first direction at least partially parallel to the radiation direction, wherein a length of the first injection-molded member in a second direction perpendicular to the first direction is greater than a length of the second injection-molded member in the second direction and a length of the antenna circuitry in the second direction, and wherein the second injection-molded member is disposed to overlap with the antenna circuitry in its entirety in the radiation direction and forms, together with the metal housing, an exterior of the portable electronic device.

2. The portable electronic device of claim 1, wherein the metal housing is positioned on an upper end and a lower end of the second injection-molded member with respect to the radiation direction of the antenna circuitry.

3. The portable electronic device of claim 1, wherein a composition ratio of a raw material and an inorganic material included in the first injection-molded member differs from that of the second injection-molded member.

4. The portable electronic device of claim 1, wherein types of a raw material and an inorganic material included in the first injection-molded member differ from those included in the second injection-molded member.

5. The portable electronic device of claim 1, wherein the first dielectric constant differs from the second dielectric constant by a dielectric constant difference of 0.5 or more.

6. The portable electronic device of claim 1, further comprising: a film surrounding at least a portion of a surface of the second injection-molded member forming the exterior of the portable electronic device.

7. The portable electronic device of claim 1, wherein the first injection-molded member forms the first organic film in a region between the at least one first part of the inner surface of the metal housing and the portion of the first injection-molded member, wherein the second injection-molded member forms the second organic film in a region between the at least one second part of the inner surface of the metal housing and the portion of the second injection-molded member, and wherein the second organic film is formed after the first organic film is formed.

8. The portable electronic device of claim 1, wherein a length measured in the second direction of a radiation area emitted in the radiation direction from the antenna circuitry is greater than a length measured in the first direction of the second injection-molded member, wherein the second direction is a direction intersecting the radiation direction and the first direction, and wherein, when the antenna circuitry is viewed from outside the portable electronic device in a direction through the second injection-molded member, the second injection-molded member overlaps the antenna circuitry in its entirety.

9. The portable electronic device of claim 1, wherein the second injection-molded member is coupled with the first injection-molded member through a coupling structure in which one of the first injection-molded member or the second injection-molded member is at least partially protruding into another one of the first injection-molded member or the second injection-molded member.

10. The portable electronic device of claim 9, wherein, in the coupling structure, the second injection-molded member protrudes into the first injection-molded member.

11. The portable electronic device of claim 9, wherein, in the coupling structure, the first injection-molded member protrudes into the second injection-molded member.

12. The portable electronic device of claim 9, wherein the coupling structure includes a shape in which the one of the first injection-molded member or the second injection-molded member penetrates into the other one of the first injection-molded member or the second injection-molded member.

13. The portable electronic device of claim 9, wherein the coupling structure includes a shape in which the one of the first injection-molded member or the second injection-molded member

penetrates into the other one of the first injection-molded member or the second injection-molded member.

14. A portable electronic device comprising: a metal housing; antenna circuitry positioned in the metal housing; a first injection-molded member including an inorganic material and having a first dielectric constant; a second injection-molded member having a second dielectric constant greater than the first dielectric constant; a first organic film disposed along and contacting at least one first part of an inner surface of the metal housing to bond with at least a portion of the first injection-molded member; and a second organic film disposed along and contacting at least one second part of the inner surface of the metal housing to bond with at least a portion of the second injection-molded member, wherein the first injection-molded member, the second injection-molded member, and the antenna circuitry are disposed along a first direction at least partially parallel to a radiation direction of the antenna circuitry, wherein a length of the first injection-molded member in a second direction perpendicular to the first direction is greater than a length of the second injection-molded member in the second direction and a length of the antenna circuitry in the second direction, and wherein the second injection-molded member is disposed to overlap with the antenna circuitry in its entirety in the radiation direction of the antenna circuitry and forms, together with the metal housing, an exterior of the portable electronic device.

15. The portable electronic device of claim 14, wherein the first injection-molded member and the second injection-molded member are positioned in the radiation direction of the antenna circuitry.

16. The portable electronic device of claim 14, wherein a composition ratio of a raw material and an inorganic material included in the first injection-molded member differs from that of the second injection-molded member.

17. The portable electronic device of claim 14, wherein types of a raw material and an inorganic material included in the first injection-molded member differ from those included in the second injection-molded member.

18. The portable electronic device of claim 14, wherein the first dielectric constant differs from the second dielectric constant by a dielectric constant difference of 0.5 or more.

19. The portable electronic device of claim 14, wherein the second injection-molded member is at least partially bonded with the first injection-molded member through a coupling structure with the first injection-molded member, and wherein, in the coupling structure, the second injection-molded member protrudes into the first injection-molded member.

20. The portable electronic device of claim 14, wherein the second injection-molded member is at least partially bonded with the first injection-molded member through a coupling structure with the first injection-molded member, and wherein, in the coupling structure, the first injection-molded member protrudes into the second injection-molded member.
